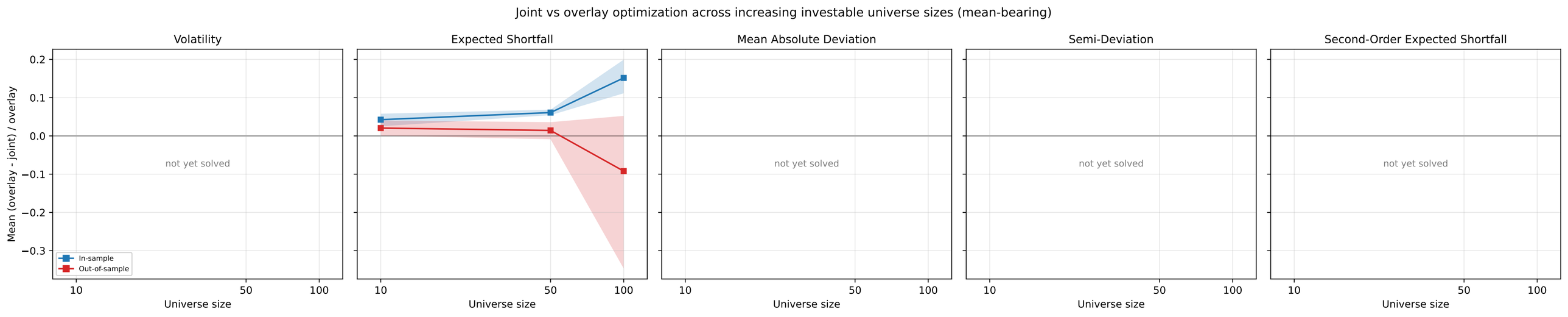
Joint vs overlay optimization across increasing investable universe sizes (mean-bearing)
Volatility
Expected Shortfall
Mean Absolute Deviation
Semi-Deviation
Second-Order Expected Shortfall
Mean (overlay - joint) / overlay
0.2
0.1
0.0
−0.1
−0.2
−0.3
not yet solved
In-sample
Out-of-sample
10
50
100
Universe size